DATA ANALYSIS AND VISUALIZATION - DAV PROJECT

COVID-19 in the United Kingdom

Infographic summary: pandemic waves, severity and inferential modelling, 2020-2026

2020-2026
WEEKLY COVID-19 TIMELINE

UK vs Poland
COMPARISON CONTEXT

2 weeks
BEST CASES-TO-DEATHS REGRESSION LAG

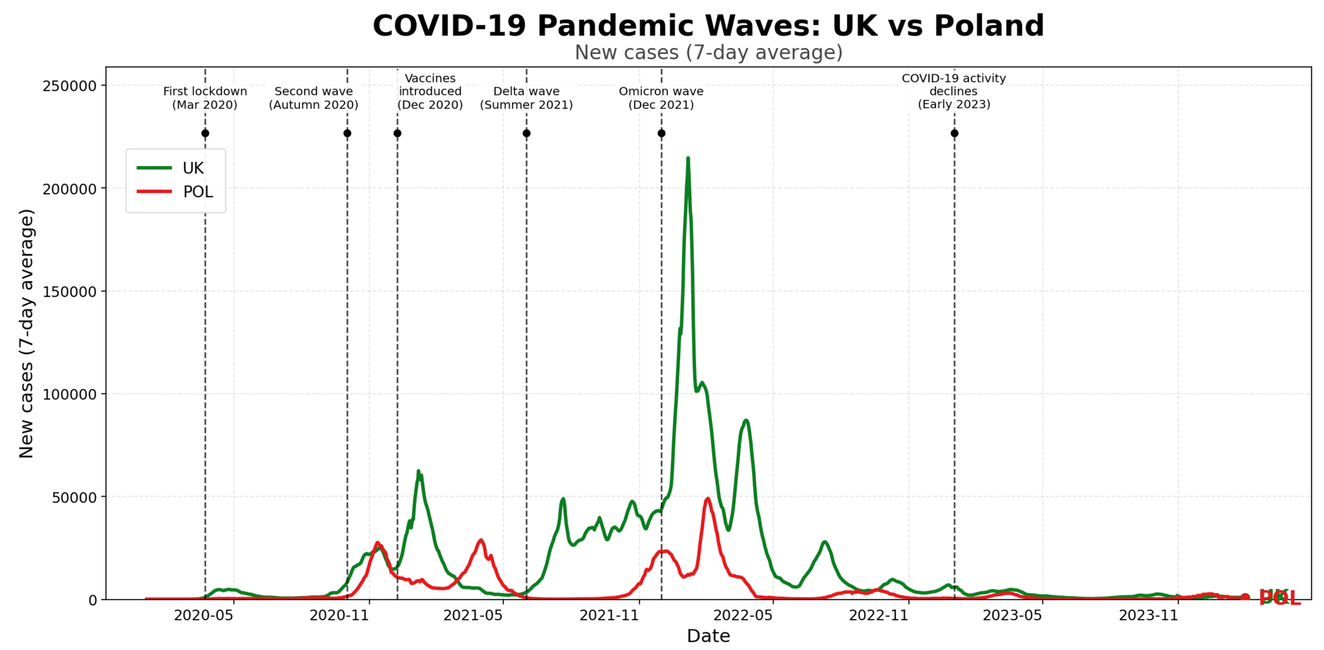
1 week
BEST HOSPITAL ADMISSIONS LAG

1. Summary statistics

2. Model evidence

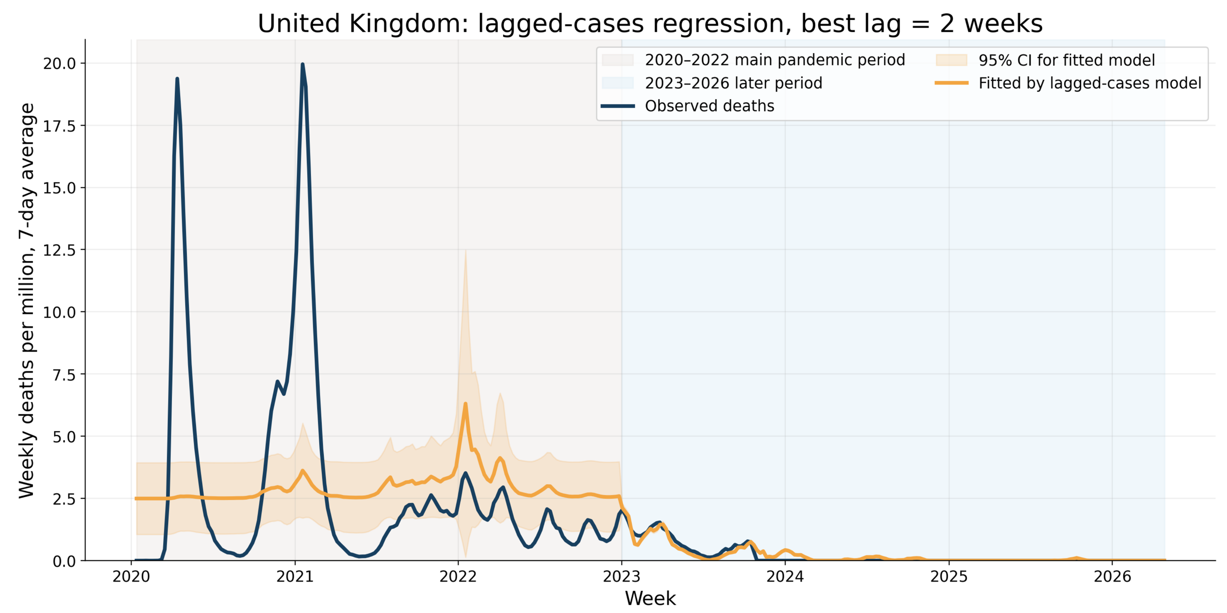
3. Inference and prediction

Pandemic waves: UK vs Poland



The UK experienced several large waves, including a major Omicron-period case peak and a strong decline after early 2023.

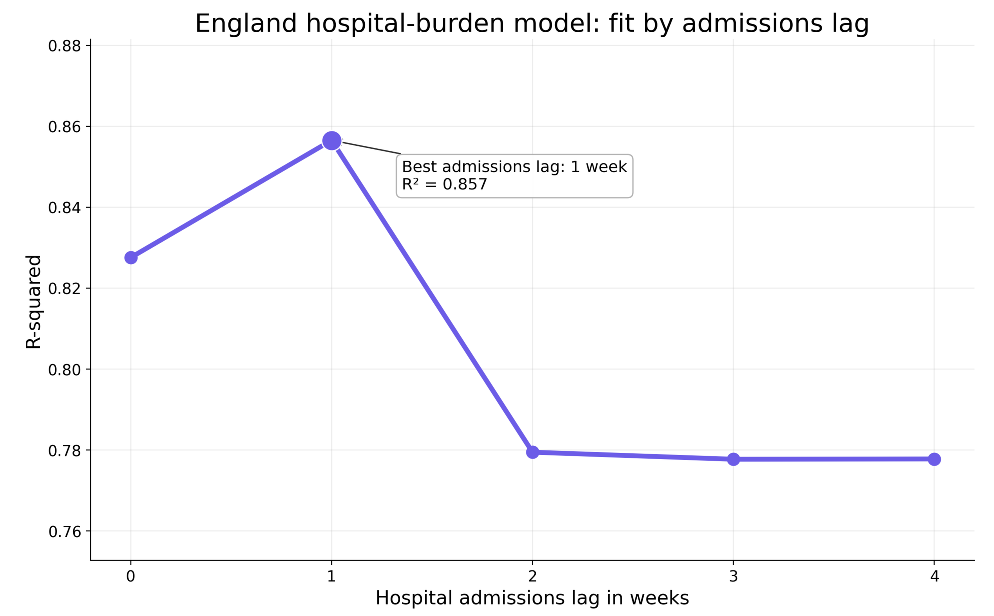
Cases-only model: limitation check



The model is intentionally shown as a limitation check; plotted fitted values are clipped at zero because deaths cannot be negative.

Lagged cases are associated with later deaths, but reported cases alone do not reproduce the largest early death waves.

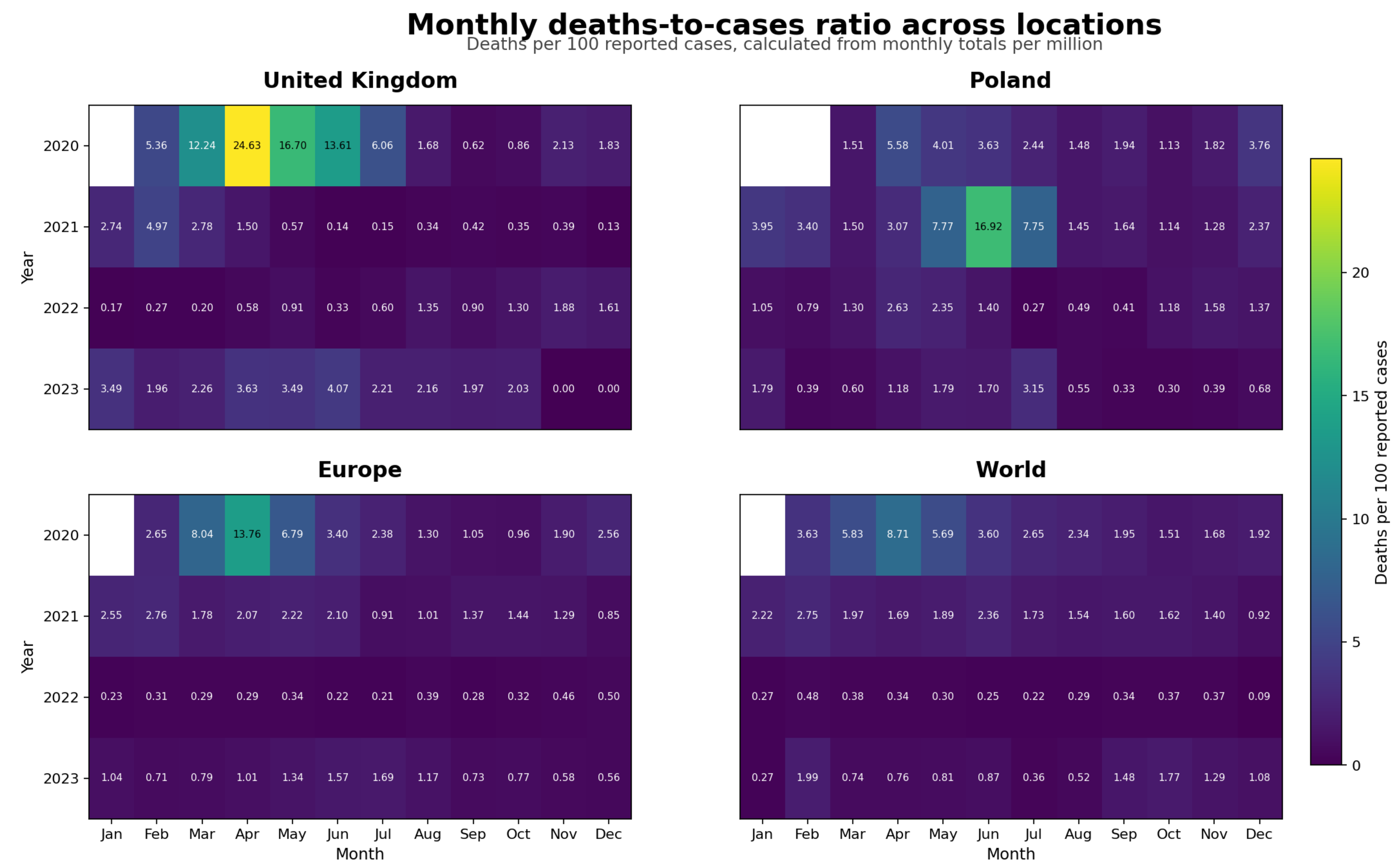
Choosing the hospital admissions lag



Model: weekly deaths ~ lagged admissions + occupied beds + PCR positivity; HAC robust standard errors used.

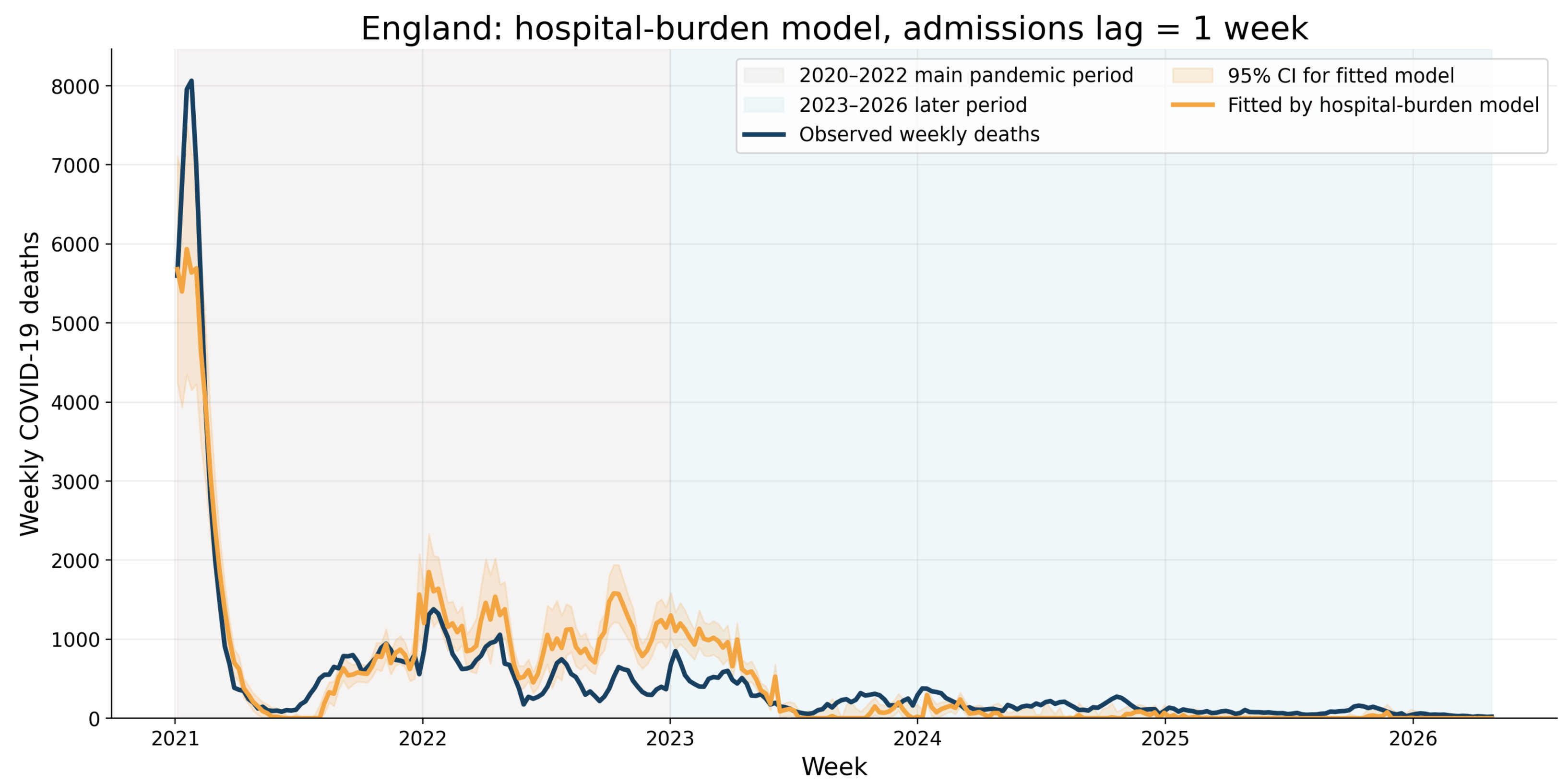
The best hospital-burden fit uses admissions lagged by one week.

Deaths-to-cases ratio by month



Larger comparative heatmap: monthly deaths per 100 reported cases across the UK, Poland, Europe and the world. The early pandemic is visibly more severe than later periods.

Hospital-burden model



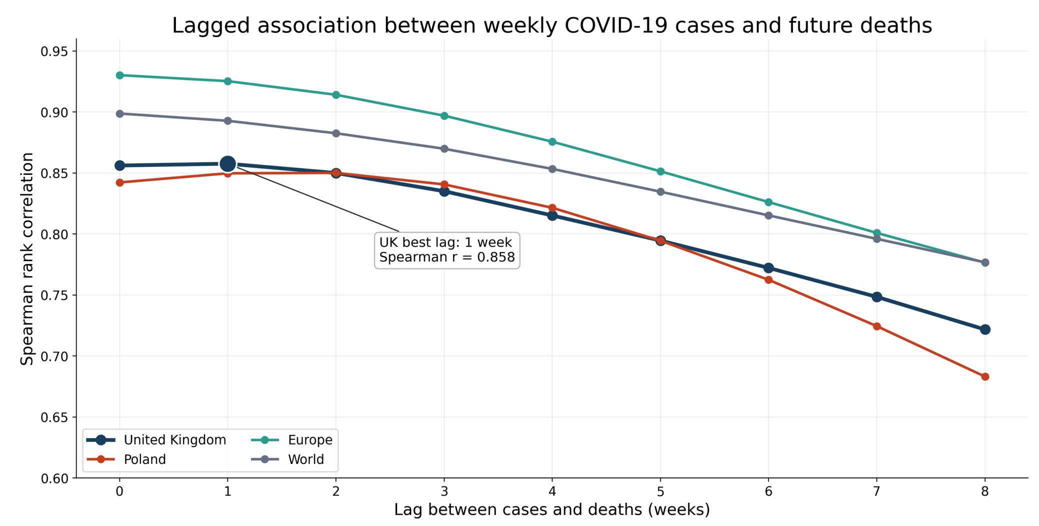
England-level UKHSA data: fitted values are for association/prediction; plotted fitted values are clipped at zero.

Hospital admissions, occupied beds and PCR positivity explain weekly deaths much better than reported cases alone.

Poster takeaway

1. **Two epidemic phases**
   Early waves and later low-death periods should not be modelled as one curve.
2. **Severity declined**
   Deaths per reported case were much higher in the first pandemic waves.
3. **Cases are not enough**
   Reported cases signal later deaths, but miss the largest early mortality peaks.
4. **Hospital burden fits better**
   Admissions lagged by one week give the strongest explanatory model.
5. **Prediction must be segmented**
   Short forecasts should use the later-period trend only.

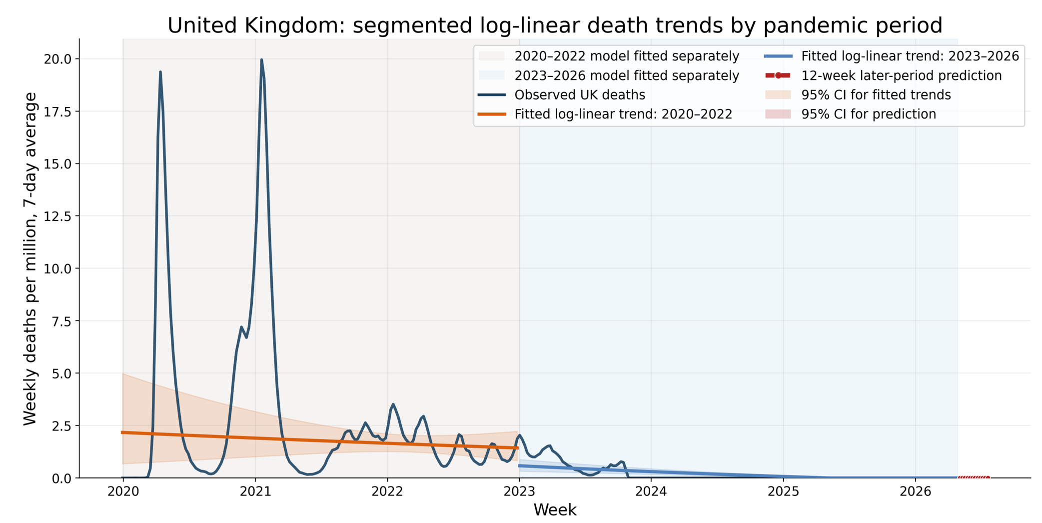
Lagged cases-deaths association



Method: Spearman rank correlation; weekly cases at week t vs deaths at week t + lag. High values indicate monotonic association, not causation.

Spearman correlation identifies the strongest UK cases-deaths association around a 1-week lag; regression fit is assessed separately.

Segmented trend prediction



Prediction uses only the later-period model. CI for the prediction is included, but is visually compressed near zero on the full pandemic scale.

Separate trends for 2020-2022 and 2023-2026 avoid one misleading trend over a declining epidemic.

Main inference conclusion

Reported cases signal future deaths, but hospital-burden indicators explain mortality more strongly. Trend prediction should be segmented by pandemic period because the epidemic declined and changed structurally.

Methods used

- Spearman lag correlation
- HAC-robust lagged regression
- Hospital-burden regression
- Segmented log-linear short prediction

Data and scripts are included in the project directory. Poster prepared as A0 PDF; plots remain readable after A4 scaling.